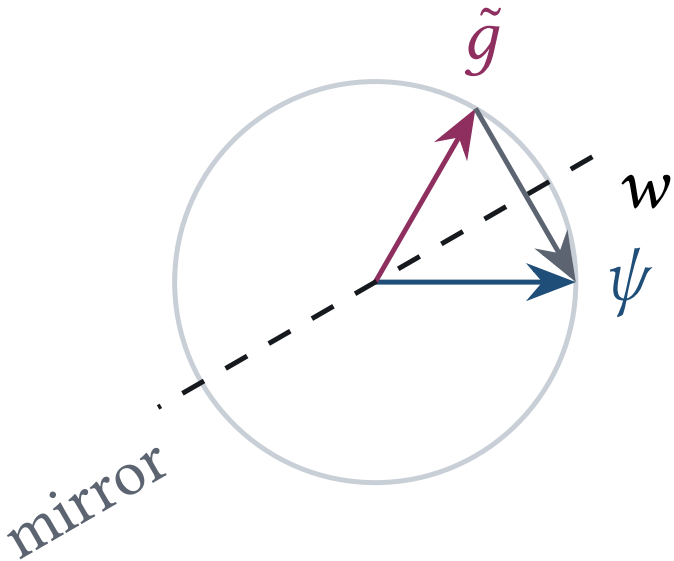


$$T = I - 2\frac{ww^\dagger}{\|w\|^2}$$

$T = I$ if $w = 0$.